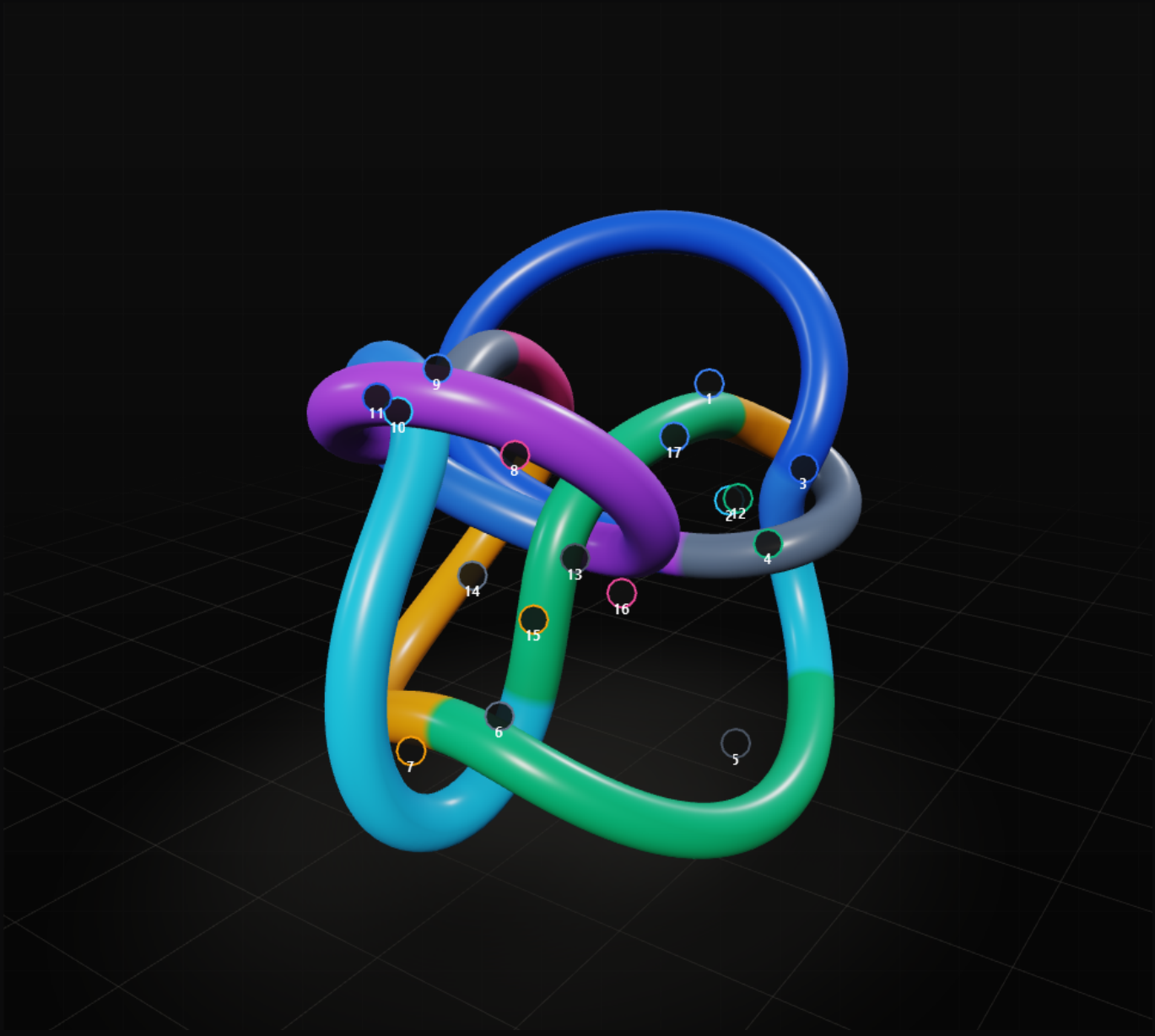
AKSIOMA

KNOTS

User Manual

An interactive 3D laboratory for knot theory



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CHAPTER 1

Getting started

AKSIOMA Knots is an interactive 3D laboratory for knot theory. You can explore well-known knots, browse a table of 12,965 prime knots, draw your own knot with the mouse, read off its invariants (writhe, crossings, Alexander and Jones polynomials) and watch it being unknotted step by step. The program is available in English and Albanian.

System requirements

- Windows, Linux or macOS with a graphics card and driver that support **OpenGL 3.3** (core profile).
- To build from source: **Qt 6.5 or newer** with CMake, or Qt Creator with a Qt 6 kit.
- A mouse with a scroll wheel is recommended for the 3D view.

Building and starting the program

If you received a ready-made `AKSIOMAKnots.exe`, simply double-click it. To build it yourself, run in the project folder:

```
cmake -S . -B build -DCMAKE_PREFIX_PATH=<path to Qt>
cmake --build build --config Release
```

Or open `CMakeLists.txt` in Qt Creator, choose a Qt 6 kit and press **Run**. On Windows, run `windeployqt` on the finished `.exe` before copying it to another computer, so that the Qt libraries travel with it.

> **Note:** The files `resources.qrc`, the `icons/` folder and `data/knottable.dat` must stay next to `CMakeLists.txt`. They contain the interface icons and the knot table.

First start

On the first start the language follows your operating system (Albanian systems get Albanian, everything else English). The window opens with the **Trefoil knot** already loaded. Every change you make in **SETTINGS** is saved automatically and restored next time.

CHAPTER 2

The main window

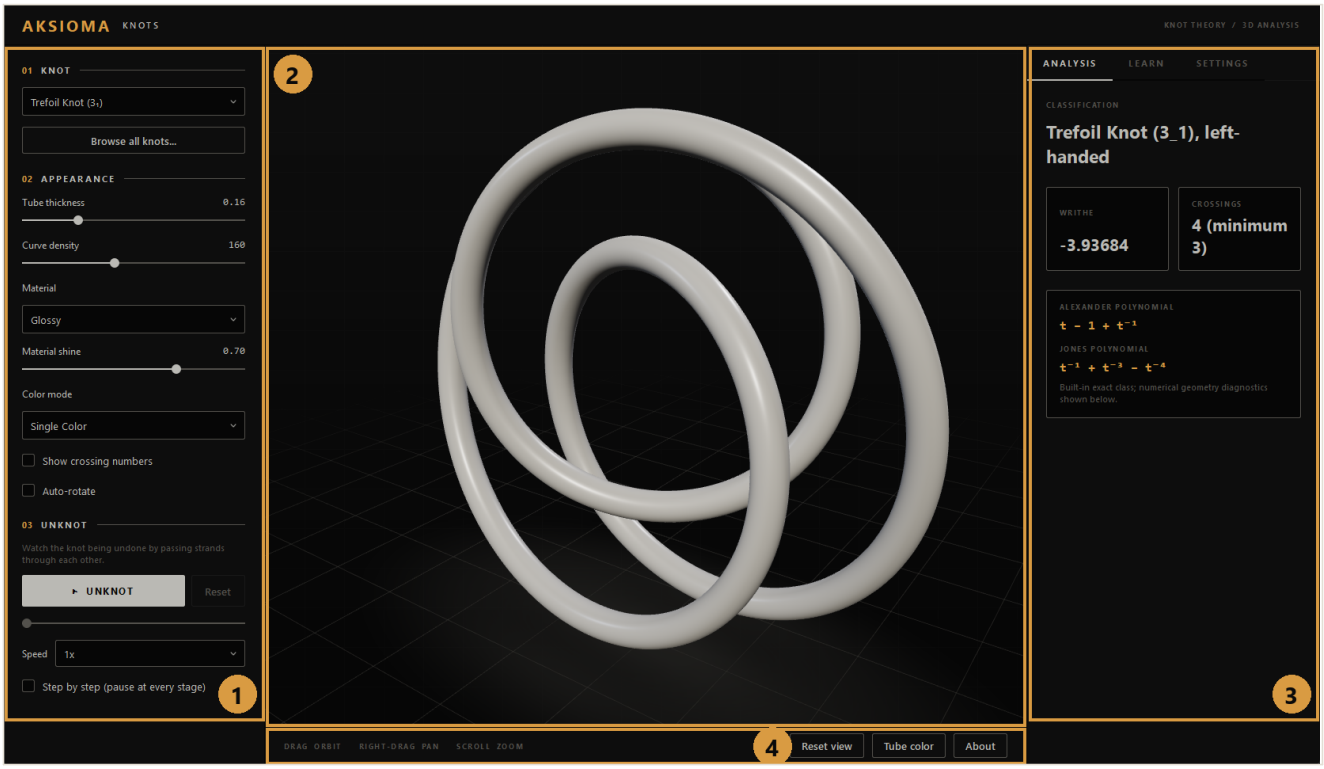


Figure 1 – The four areas of the main window.

1 Control panel	Three numbered sections: **01 KNOT** (choose what to show), **02 APPEARANCE** (how it looks) and **03 UNKNOT** (the unknotting animation). The panel scrolls if it does not fit the window.
2 3D view	The knot drawn as a lit 3D tube above a floor grid. You can turn, move and zoom it freely.
3 Information tabs	**ANALYSIS** shows the computed invariants, **LEARN** explains them in plain words, **SETTINGS** holds language, theme and window options.
4 Bottom bar	A reminder of the mouse controls and the buttons **Reset view**, **Tube color** and **About**.

The borders between the three columns can be dragged to make the 3D view wider or narrower.

CHAPTER 3

Moving around and choosing a knot

Controlling the 3D view

Left-drag	Orbit – rotate the knot around its centre.
Right-drag	Pan – slide the knot sideways or up and down.
Scroll wheel	Zoom in and out.
Reset view	Return the camera to the starting position.
Tube color	Pick the base colour of the tube (used by the **Single Color** mode).
F11 / Esc	F11 switches full screen on and off; Esc leaves full screen.

Built-in knots (section 01 KNOT)

Open the drop-down list at the top of the control panel and choose one of the presets:

- **Unknot** – a plain closed loop, the trivial knot (0_1).
- **Trefoil Knot (3_1)**, **Figure-Eight Knot (4_1)**, **Cinquefoil (5_1)**, **Stevedore Knot (6_1)** – the classic small knots.
- **DNA Double Helix** and **Celtic Armor** – multi-strand *links* (several closed loops linked together). For these the program shows the Gauss linking numbers.
- **Custom Torus T(p,q)**, **Custom Curve** and **Knot Table** – described below and on the next pages.

Custom torus knot T(p,q)

A torus knot winds **p** times around the axis of a doughnut and **q** times through its hole. Choose **Custom Torus T(p,q)** and type two whole numbers from 1 to 20 into the **p** and **q** fields; the knot is rebuilt at once.

If p and q share a common factor, the result is a *torus link* with several components, e.g. T(4,2) has 2 components. The figure shows T(5,3) with **Strand Colors** and **Show crossing numbers** turned on.

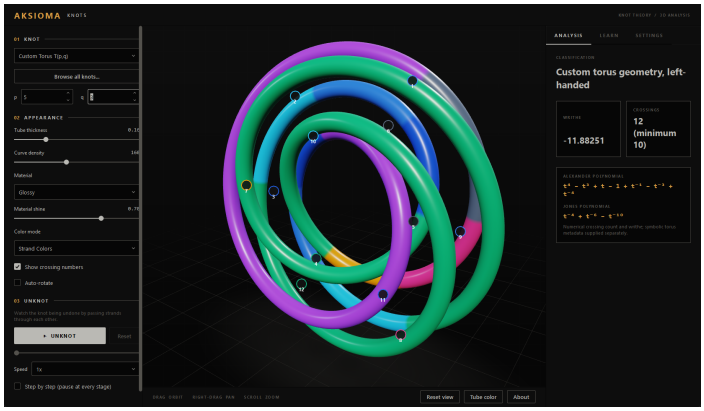


Figure 2 – The torus knot T(5,3).

CHAPTER 4

The Knot Browser

Click **Browse all knots…** (under the knot list) to open the Knot Browser. It contains all **12,965 prime knots with up to 13 crossings** from the KnotInfo database, each with its 3D shape and invariants.

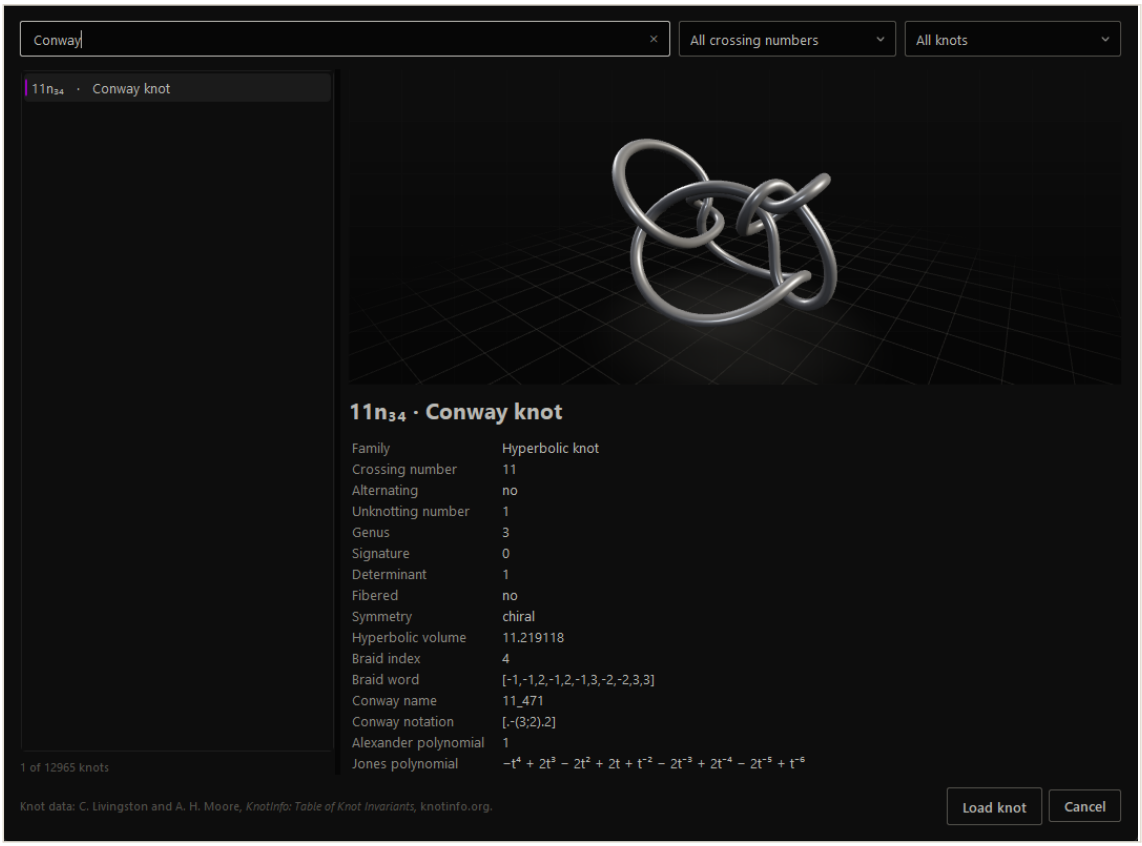


Figure 3 – The Knot Browser, searching for the Conway knot.

- **Search box** – type a knot name such as `8_19` or `11n34`, or a word such as *trefoil* or *Conway*. The little **×** clears it.
- **Crossing filter** – **All crossing numbers** or only knots with a given number of crossings.
- **Type filter** – **All knots**, **Alternating**, **Non-alternating**, **Fibered** or **Amphicheiral**.
- **List** – click a knot to see its 3D preview and properties on the right; the count below the list (e.g. *1 of 12965 knots*) shows how many match.
- **Load knot** – shows the selected knot in the main window (double-clicking a row does the same). **Cancel** closes the browser without changes.

> **Tip:** The properties shown include family, crossing number, unknotting number, genus, signature, determinant, symmetry, hyperbolic volume, braid index and word, Conway notation, and both polynomials. See Chapter 8 for what they mean.

CHAPTER 5

Your own knots: formulas, points and drawing

Choose **Custom Curve** in the knot list. A text box appears with two buttons, **Draw a knot...** and **APPLY**. You can describe the curve in two ways:

1. Formulas in t

Write three formulas separated by semicolons. The parameter `t` runs from 0 to 2π, and the curve must close up (end where it starts):

```
x=(2+cos(2*t))*cos(3*t);
y=(2+cos(2*t))*sin(3*t);
z=sin(4*t);
```

Allowed: `+ − * / ^`, brackets, `pi`, and the functions sin, cos, tan, asin, acos, atan, sinh, cosh, tanh, exp, log/ln, sqrt, abs, floor, ceil.

2. A list of points

One point per line as `x, y, z` (at least three points). The last point is joined back to the first. Press **APPLY** to build the curve.

3. Drawing a knot with the mouse

Click **Draw a knot...**. Hold the **left mouse button** and draw one closed loop that crosses itself. Crossings are found automatically and marked with small circles; **click a circle** to swap which strand goes over. **Alternate** makes the crossings go over, under, over, under along the curve; **Mirror** swaps every crossing; **Clear** starts again. Press **USE THIS KNOT** to send it to the 3D view.

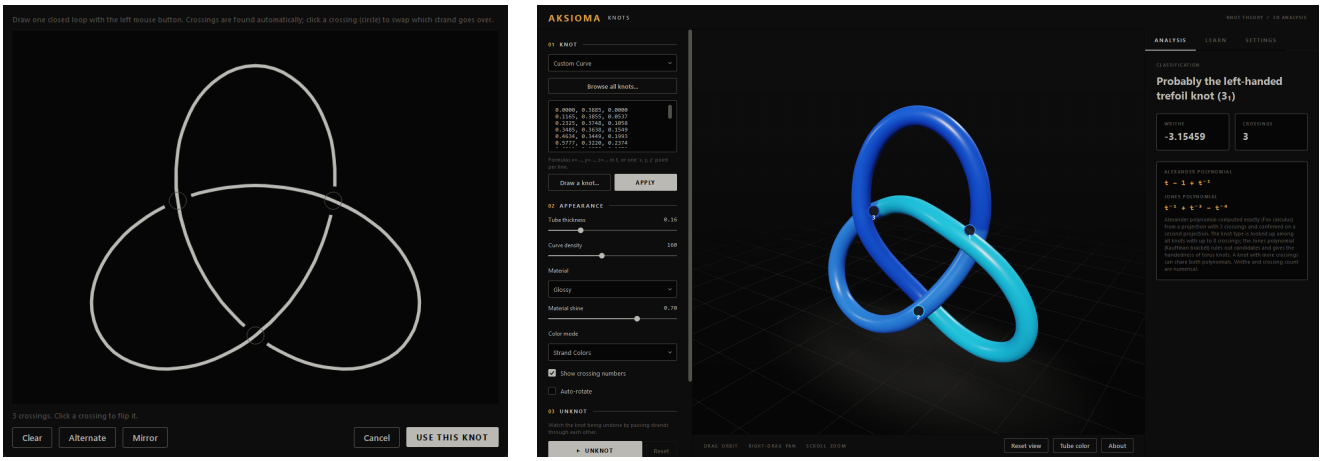


Figure 4 – Left: a hand-drawn loop with 3 crossings. Right: the result in 3D – the program identifies it as the left-handed trefoil knot.

CHAPTER 6

Appearance

Section **02 APPEARANCE** changes only how the knot looks, never the mathematics.

Tube thickness	Radius of the tube. If strands would touch, the program thins the tube automatically and tells you so.
Curve density	Number of points along the curve. Higher values give a smoother tube but need more graphics power.
Material	**Glossy**, **Satin**, **Metal** or **Rope**.
Material shine	How strong the highlights are.
Color mode	**Single Color** – the whole tube in the colour chosen with **Tube color**. **Crossing Colors** – each crossing gets its own colour, shared by its two branches. **Strand Colors** – each arc between two under-crossings gets its own colour, so the path of the knot is easy to follow.
Show crossing numbers	Numbered markers on every crossing found in the current (XY) projection.
Auto-rotate	The knot turns slowly by itself – useful for presentations.

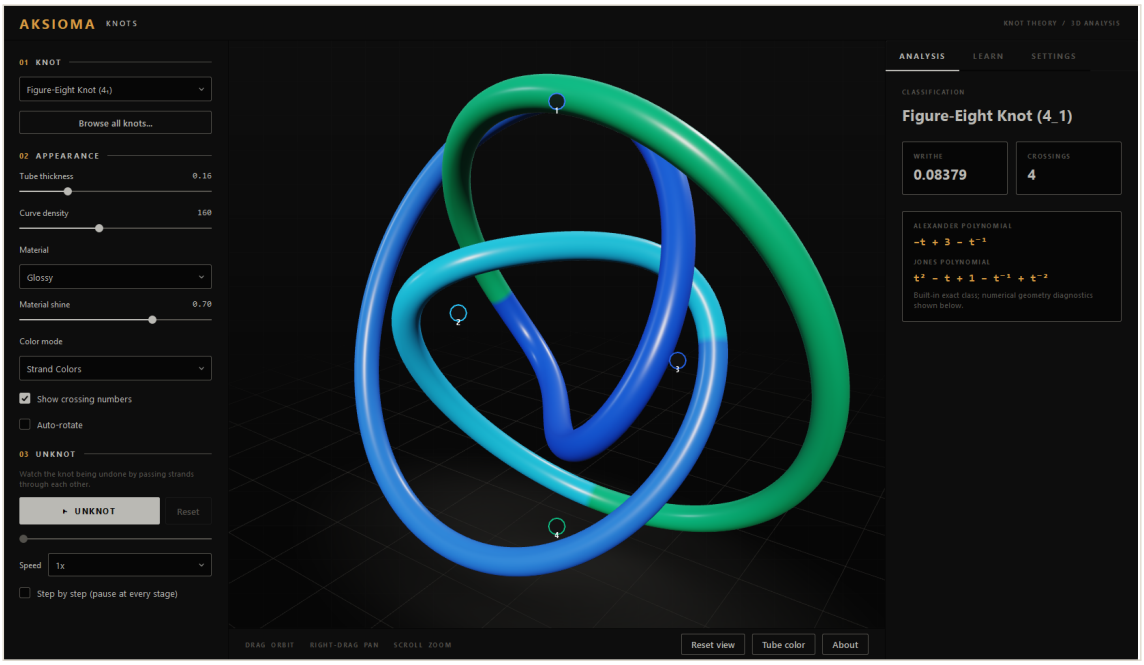


Figure 5 – Figure-eight knot with Strand Colors and crossing numbers.

CHAPTER 7

The unknotting animation

Section **03 UNKNOT** shows how a knot can be turned into a plain circle if strands are allowed to pass through each other. It works for any single closed curve (not for multi-strand links).

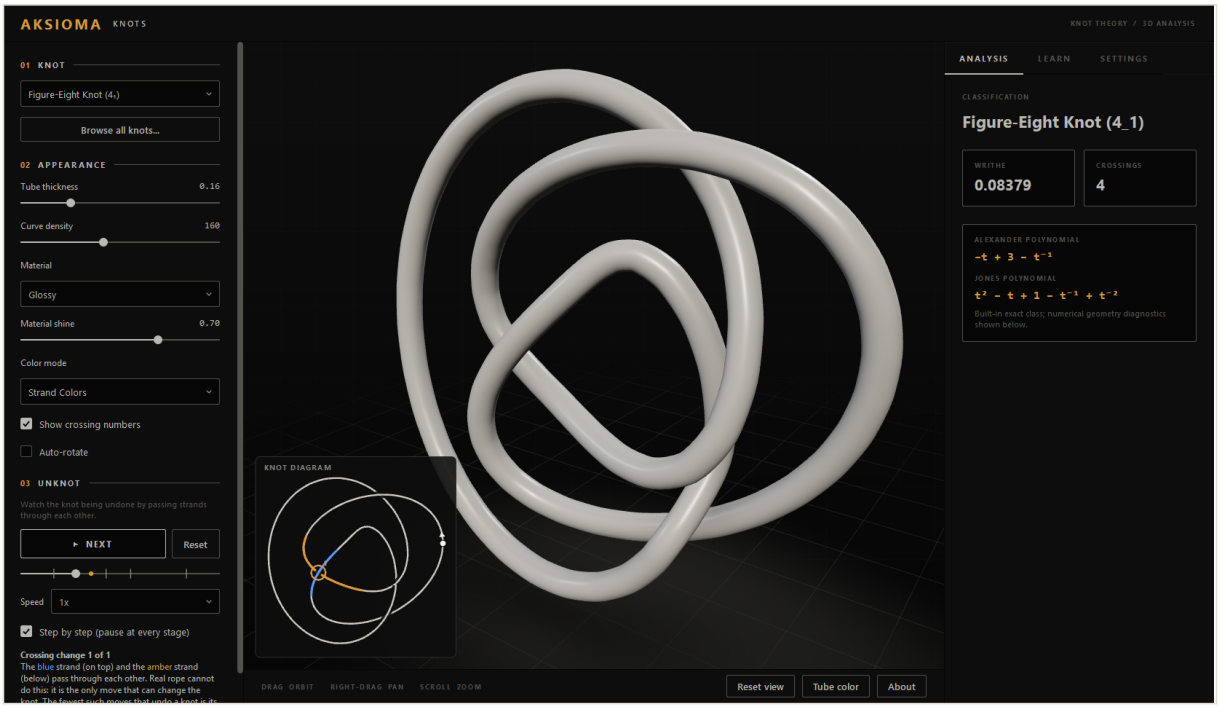


Figure 6 – A crossing change: the blue strand (on top) and the amber strand pass through each other. The knot diagram is shown at the lower left.

▶ UNKNOT / PAUSE / PLAY	Start, pause and continue the animation.
Reset	Return to the original knot.
Timeline slider	Drag to jump to any moment. The tick marks show the stages.
Speed	0.25x, 0.5x, 1x, 2x or 4x.
Step by step	Pause at every stage; press **▶ NEXT** to continue. Recommended for teaching.

The stages are: **Choosing a viewpoint** → **A knot diagram** → **Crossing change 1 of n** → **Downhill all the way** → **Pulling it open** → **A plain circle**. At the end the program reports how many crossing changes were needed – an upper bound for the knot's unknotting number.

CHAPTER 8

The information tabs

The **ANALYSIS** tab is updated every time the knot changes:

- **CLASSIFICATION** – the name of the knot, and for chiral knots whether it is left- or right-handed. For your own curves the program compares the polynomials with all knots up to 8 crossings and writes e.g. *Probably the left-handed trefoil knot*.
- **WRITHE** – how much the 3D curve coils in space. It depends on the shape, so the same knot drawn differently can have a different writhe.
- **CROSSINGS** – crossings seen in the current picture; *(minimum n)* gives the true crossing number when it is known.
- **ALEXANDER POLYNOMIAL** – a classical invariant, computed exactly. It cannot tell a knot from its mirror image.
- **JONES POLYNOMIAL** – a stronger invariant (1984) that usually does tell mirror images apart.
- The small grey note at the bottom says how each value was obtained (from the knot table, computed exactly, or numerically).

The **LEARN** tab (right) explains these concepts in plain language for the knot on screen, and describes the colour mode you are using.

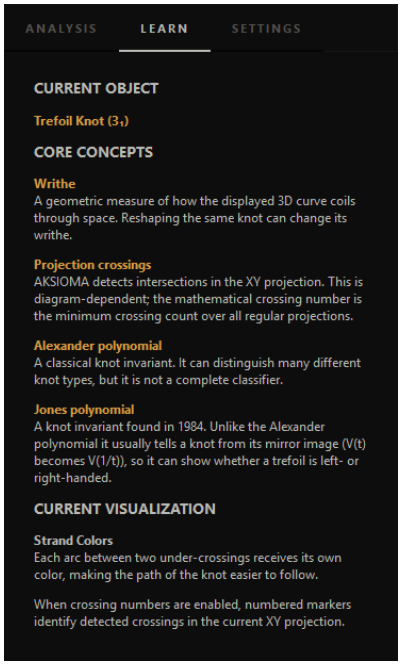


Figure 7 – The LEARN tab.

> **Important:** Different knots always have different invariants only in one direction: if two polynomials differ, the knots are different – but equal polynomials do not prove that two knots are the same. That is why the program says *Probably*.

Settings and About

Open the **SETTINGS** tab. Changes are saved automatically.

- **LANGUAGE** – *English* or *Shqip*. The window is rebuilt in the new language and your knot and view are kept.
- **Theme** – **Dark** or **Light**.
- **Text size** – **Small**, **Normal**, **Large** or **Extra large**.
- **Full screen (F11)** – switch full screen now.
- **Start in full screen** – always open full screen.
- **Remember window size** – reopen with the last size and position.
- **Restore defaults** – reset every setting (asks for confirmation).

About

The **About** button shows the version, the licence (GNU GPL v3), the source of the knot data (KnotInfo, C. Livingston and A. H. Moore) and the mathematical methods used. **About Qt** shows the Qt licence.

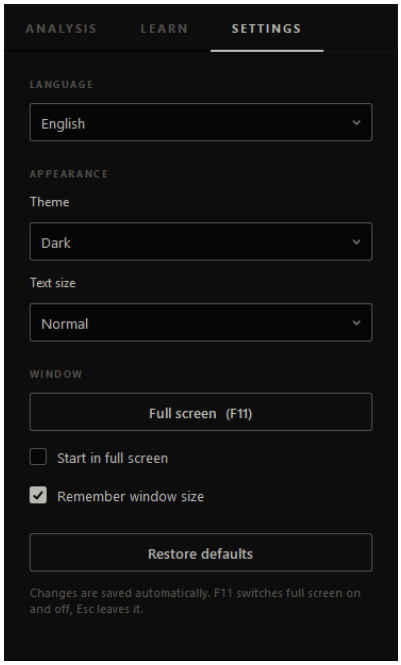


Figure 8 – The SETTINGS tab.

CHAPTER 9

Troubleshooting and glossary

Problem	Solution
The 3D view is black or the program closes at once	Your graphics driver must support OpenGL 3.3. Update the graphics driver.
Missing DLL error on Windows	Run `windeployqt` on the .exe, or start it from Qt Creator.
"The knot table is missing / damaged"	Rebuild so that `data/knottable.dat` is included.
"The requested curve could not be generated"	Check that formulas define x, y and z, that each point has exactly three numbers, and that there are at least three points.
"Unknotting needs a single closed curve"	The DNA and Celtic presets are links; choose a single knot.
Drop-down lists open behind the window in full screen	Press Esc to leave full screen, then F11 again.

Glossary

Knot	A closed loop in space that does not cross itself.
Unknot	A loop that can be deformed into a plain circle.
Link	Several closed loops, possibly tangled together.
Crossing number	The smallest number of crossings in any picture of the knot.
Unknotting number	The fewest crossing changes needed to turn the knot into the unknot.
Mirror image / chiral	The reflected knot; a knot is chiral if it differs from its mirror image.
Amphicheiral	Equal to its own mirror image (e.g. the figure-eight knot).
Alternating	Has a picture where crossings go over, under, over, under all the way round.
Torus knot T(p,q)	A knot that lies on the surface of a doughnut.